

Practice : Divide and conquer

Deadline: May.6/7/8. Must be checked onsite (during labs).Online checks via email or messaging apps are not accepted.

1. Objective

This Practice has one question based on **Chapter 5.3 Counting Inversions** in the lecture. It is intended to help students understand the basic concepts of Divide and Conquer.

2. Task

You need finish this task by two ways:

Solution	Time Complexity	Space Complexity
Divide and Conquer	$O(n \log n)$	$O(n)$ need temporary array
Brute Force	$O(n^2)$	$O(1)$

Description:

Find the number of pairs in a sequence where a larger number appears before a smaller number.

Input Description:

The first line contains an integer **n**, representing the number of elements in the sequence.

The second line contains n integers separated by spaces.

Output Description:

Output a single line representing the number of pairs where a larger number appears before a smaller number.

Input Test case1:

```
12
1 5 4 8 10 2 6 9 12 11 3 7
```

Output1

```
22
```

Input Test case2:

```
10
9 8 3 2 4 8 3 2 4 5
```

Output2

```
25
```

Input Test case3:

In the File `1.in`.

Output Test case3:

```
100660982
```

In this section, you can try the command:

Java

1. Mac or Linux

```
time java Main < 1.in
```

2. Windows power shell

```
Measure-Command { cmd /c "java Main < 1.in" }
or
(Measure-Command { cmd /c "java Main < 1.in" }).TotalMilliseconds
```

C++

1. Mac or Linux

```
time ./Main < 1.in
```

2. Windows power shell

```
Measure-Command { cmd /c ".\Main.exe < 1.in" }
or
(Measure-Command { cmd /c ".\Main.exe < 1.in" }).TotalMilliseconds
```

4. Submission

Show the following to your tutor or SA **onsite**:

Grading:

- Submissions on time:
 - 0.5 points for first two test cases by using **Divide and conquer** solution.
 - 0.2 points for the correct answer of File `1.in` by using **Divide and conquer** solution.
 - 0.1 point for space complexity with $O(n)$ by using **Divide and conquer** solution.
 - 0.2 points for a comparison of the runtime between the **brute-force** algorithm and the **divide and conquer** algorithm. With **fast input being used** in both cases, record the time between Brute-force and divide and conquer algorithm.

For example:

Times	Brute-force	Divide-and-conquer
First time	489	62
Second time	385	41

- Late submissions: 0 points.